

1. GAN Architecture

Explain the adversarial process in GAN training. What are the goals of the generator and discriminator, and how do they improve through competition? Diagram of the GAN architecture showing the data flow and objectives of each component.

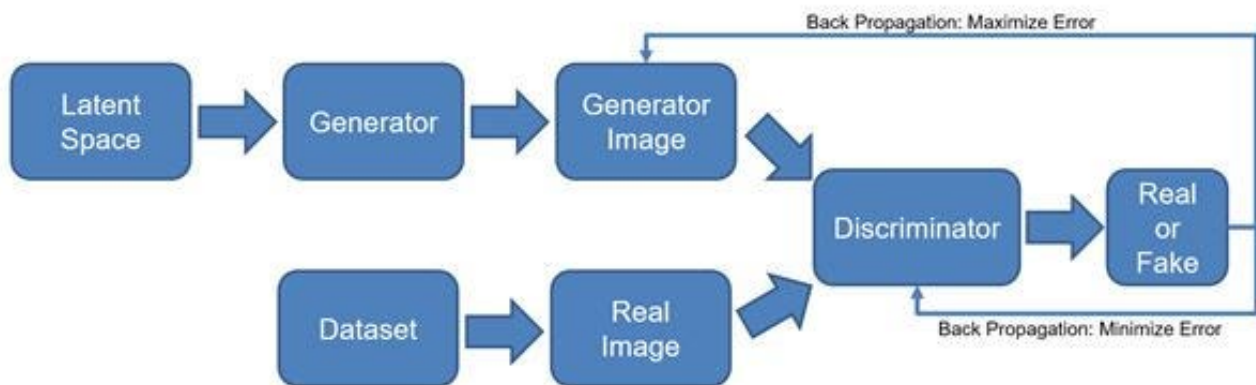
Ans:

Adversarial Process in GAN Training:

- The **generator** tries to create fake data that looks real.
- The **discriminator** tries to tell apart real data from fake data.
- They compete:
 - The **generator** improves by trying to fool the discriminator.
 - The **discriminator** improves by getting better at detecting fakes.
- Training continues until the generator makes data so real that the discriminator cannot easily tell the difference (50/50 chance).

Goals:

- **Generator's Goal:** Maximize the discriminator's mistakes (make fake data that looks real).
- **Discriminator's Goal:** Minimize its classification error (correctly identify real vs fake).



2. Ethics and AI Harm

Choose one of the following real-world AI harms discussed in Chapter 12:

- Representational harm
- Allocational harm
- Misinformation in generative AI

Describe a real or hypothetical application where this harm may occur. Then, suggest **two harm mitigation strategies** that could reduce its impact based on the lecture.

Ans:

Chosen Harm:

→ Misinformation in Generative AI

Application where this harm may occur:

→ A generative AI chatbot that provides health advice may generate false medical information, leading users to take unsafe actions.

Two Harm Mitigation Strategies:

1. Human-in-the-Loop Review:

Ensure AI outputs, especially for sensitive domains like healthcare, are reviewed by human experts before reaching users.

2. Fact-Checking Systems:

Integrate real-time verification tools that cross-check AI-generated content against trusted knowledge bases.

5. Legal and Ethical Implications of GenAI

Discuss the legal and ethical concerns of AI-generated content based on the examples of:

- Memorizing private data (e.g., names in GPT-2)
- Generating copyrighted material (e.g., Harry Potter text)

Do you believe generative AI models should be restricted from certain data during training? Justify your answer.

Ans:

Memorizing Private Data:

Models like GPT-2 have shown that AI can memorize and unintentionally reproduce private information such as names, emails, or other sensitive data from training datasets. This raises serious privacy concerns because individuals' data could be exposed without consent.

Generating Copyrighted Material:

Generative models have also produced copyrighted content, like Harry Potter paragraphs, almost word-for-word. This creates legal risks related to intellectual property violations and copyright infringement, since the AI is recreating protected material without permission.

Should AI models be restricted from certain data during training?

Yes, I believe generative AI models should be restricted from using private or copyrighted data during training.

- Justification:
 - Protects individuals' privacy and prevents leakage of sensitive information.
 - Respects intellectual property laws and encourages fair use.
 - Ensures ethical AI development that minimizes potential harm to people and creators.

6. Bias & Fairness Tools

Visit [Aequitas Bias Audit Tool](#).

Choose a bias metric (e.g., false negative rate parity) and describe:

- What the metric measures
- Why it's important
- How a model might fail this metric

Optional: Try applying the tool to any small dataset or use demo data.

Ans:

Chosen Metric:

→ False Negative Rate Parity

What the Metric Measures:

False Negative Rate Parity measures whether different groups (e.g., race, gender) have similar false negative rates — meaning how often the model incorrectly predicts a "no" (negative) when the true label should be "yes" (positive).

Why It's Important:

If one group experiences a higher false negative rate, they may be unfairly denied opportunities or resources. For example, in a loan approval model, if a certain group receives more false negatives, they are unfairly rejected for loans even though they qualify.

How a Model Might Fail This Metric:

A model might fail if it is trained on biased or imbalanced data, leading it to wrongly reject more qualified individuals from certain groups compared to others.

Optional Trial with Demo Data:

Using Aequitas' demo data, we can see different bias metrics for a sample recidivism dataset, where certain racial groups had higher false negative rates than others, indicating unfairness.